

120 Days of Quantitative Finance

Month 5: Advanced Quant Systems

Week 17: Volatility & Risk

Day 115: Risk-Neutral vs Real-World Measure

1 Introduction

In quantitative finance, two probability measures are fundamental:

- Real-world measure ($\mathbb{P}$)
- Risk-neutral measure ($\mathbb{Q}$)

Understanding their difference is essential for distinguishing between forecasting and pricing.

2 Real-World Measure ($\mathbb{P}$)

Under the real-world measure:

$$dS_t = \mu S_t dt + \sigma S_t dW_t^{\mathbb{P}}$$

- μ : expected return (includes risk premium)
- Used for forecasting returns

3 Risk-Neutral Measure ($\mathbb{Q}$)

Under the risk-neutral measure:

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}$$

- Drift becomes the risk-free rate r
- Investors are indifferent to risk

4 Key Principle: No-Arbitrage

The fundamental pricing equation:

$$\text{Price} = e^{-rT} \mathbb{E}^{\mathbb{Q}}[\text{Payoff}]$$

5 Why Change Measure?

- Simplifies pricing
- Eliminates risk premia
- Ensures no-arbitrage consistency

6 Interpretation

- $\mathbb{P}$: how the world actually evolves
- $\mathbb{Q}$: how assets must be priced

7 Connection to Volatility

- GARCH estimates volatility under $\mathbb{P}$
- Implied volatility reflects $\mathbb{Q}$

8 Market Price of Risk

The difference between $\mathbb{P}$ and $\mathbb{Q}$ is captured by:

$$\lambda = \frac{\mu - r}{\sigma}$$

9 Applications

- option pricing
- risk-neutral valuation
- derivative modeling

10 Conclusion

The distinction between $\mathbb{P}$ and $\mathbb{Q}$ is central to quantitative finance, separating prediction from pricing.